

Exercise (FNM-CPA implies IND-CPA). Let $\mathfrak{C} = (\text{Gen}, \text{Enc}, \text{Dec})$ be a public key cryptosystem and let $\pi : \mathcal{M} \times \mathcal{M}^n \rightarrow \{0, 1\}$ be t_π -time predicate. We say that the crypto-system is (t, t_m, ε) -non-malleable with respect to the relation π (FNM-CPA) if for all t -time adversaries $\mathcal{A}$ the advantage

$$\text{Adv}_{\mathfrak{C}, \pi}^{\text{f-nm-cpa}}(\mathcal{A}) = |\Pr [\mathcal{G}_0^{\mathcal{A}} = 1] - \Pr [\mathcal{G}_1^{\mathcal{A}} = 1]| \leq \varepsilon$$

where the security games are defined as follows

$\mathcal{G}_0^{\mathcal{A}}$ <pre> [(sk, pk) ← Gen $\mathcal{M}_0 \leftarrow \mathcal{A}(\text{pk})$ $m \leftarrow \mathcal{M}_0$ $c \leftarrow \text{Enc}_{\text{pk}}(m)$ $\hat{c}_1, \dots, \hat{c}_n \leftarrow \mathcal{A}(c)$ if $c \in \{\hat{c}_1, \dots, \hat{c}_n\}$ then return 0 return $\pi(m, \text{Dec}_{\text{sk}}(\hat{c}_1), \dots)$ </pre>	$\mathcal{G}_1^{\mathcal{A}}$ <pre> [(sk, pk) ← Gen $\mathcal{M}_0 \leftarrow \mathcal{A}(\text{pk})$ $m \leftarrow \mathcal{M}_0, \bar{m} \leftarrow \mathcal{M}_0$ $\bar{c} \leftarrow \text{Enc}_{\text{pk}}(\bar{m})$ $\hat{c}_1, \dots, \hat{c}_n \leftarrow \mathcal{A}(\bar{c})$ if $c \in \{\hat{c}_1, \dots, \hat{c}_n\}$ then return 0 return $\pi(m, \text{Dec}_{\text{sk}}(\hat{c}_1), \dots)$ </pre>
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and the sampling algorithm $\mathcal{M}_0$ is guaranteed to be a t_m -time algorithm. Analyse which properties the target relation π must satisfy in order to be able to prove that FNM-CPA security implies IND-CPA. Derive a quantitative measure, which shows how effective is the relation π in this reduction. Interpret the results.

Solution.